

BATTLE CREEK KELLOGG, MI, USA

WMO: 725396

Lat: 42.308N

Lon: 85.251W

Elev: 283

StdP: 97.97

Time Zone: -5.00 (NAE)

Period: 98-19

WBAN: 14815

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.1	-14.1	-21.1	0.6	-15.9	-18.3	0.8	-13.3	11.6	-5.3	10.8	-3.7	3.3	210	0.544

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.7	32.0	22.9	30.3	22.0	28.8	21.3	24.5	29.4	23.6	28.2	22.8	27.1	5.2	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.9	18.2	27.1	22.2	17.5	26.4	21.3	16.5	25.4	75.3	29.5	71.8	28.2	68.3	27.0	28.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.6	9.1	8.2	-21.1	34.0	3.9	1.7	-23.9	35.2	-26.2	36.2	-28.4	37.1	-31.3	38.3
			-21.4	26.2	3.8	1.1	-24.1	27.0	-26.4	27.6	-28.5	28.2	-31.3	28.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	9.8	-4.3	-3.2	2.4	9.3	15.5	20.4	22.5	21.5	17.9	11.1	4.6	-1.2	(6)
(7)		DBStd	10.69	6.08	5.81	6.34	5.19	4.85	3.81	3.04	2.86	4.18	4.84	5.10	5.59	(7)
(8)		HDD10.0	1719	444	370	252	74	9	0	0	0	1	45	174	349	(8)
(9)		HDD18.3	3527	702	602	496	275	114	22	3	6	59	230	411	606	(9)
(10)		CDD10.0	1639	1	1	15	52	180	313	389	358	237	79	13	2	(10)
(11)		CDD18.3	406	0	0	1	3	28	85	134	106	44	6	0	0	(11)
(12)		CDH23.3	3327	0	0	7	28	247	708	1170	772	353	41	0	0	(12)
(13)		CDH26.7	956	0	0	1	3	59	217	383	192	95	6	0	0	(13)
(14)	Wind	WSAvg	3.9	4.5	4.4	4.3	4.5	3.9	3.5	3.2	3.0	3.2	3.8	4.3	4.4	(14)
(15)	Precipitation	PrecAvg	902	54	43	50	76	100	95	87	91	95	88	72	51	(15)
(16)		PrecMax	1112	108	104	106	150	234	183	156	168	308	181	166	85	(16)
(17)		PrecMin	720	18	10	12	12	35	40	15	19	38	19	17	16	(17)
(18)		PrecStd	125	28	25	25	34	54	41	41	41	60	44	41	20	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.2	15.5	24.0	26.6	30.9	33.1	34.5	32.7	31.8	27.6	20.6	16.1	(19)
(20)		MCWB	11.7	11.3	16.8	16.6	21.0	23.1	23.9	23.4	21.7	20.9	14.3	14.1	(20)	
(21)		2%	DB	9.4	11.4	19.1	23.2	28.3	31.1	32.2	30.2	29.4	24.0	17.2	12.3	(21)
(22)		MCWB	7.3	8.9	13.8	15.2	19.3	22.0	23.6	22.6	21.6	17.7	12.8	10.4	(22)	
(23)		5%	DB	6.3	7.5	15.6	21.0	26.3	29.1	30.4	28.7	27.3	21.4	14.8	8.9	(23)
(24)		MCWB	4.4	5.1	11.2	13.8	18.5	21.2	22.3	21.4	20.1	16.5	11.3	7.1	(24)	
(25)		10%	DB	3.7	4.5	12.2	18.2	23.9	27.4	28.7	27.4	25.1	19.1	12.6	6.2	(25)
(26)		MCWB	2.4	2.4	8.4	12.0	17.3	20.5	21.6	20.9	18.9	15.0	9.4	4.4	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.1	13.0	17.5	19.1	22.4	24.7	26.1	25.3	24.1	21.7	15.7	14.3	(27)
(28)		MDCB	13.0	14.3	24.3	24.1	28.2	30.9	31.5	30.6	29.4	26.5	18.5	16.3	(28)	
(29)		2%	WB	7.9	9.0	14.7	16.7	20.8	23.4	24.7	24.0	22.8	19.5	13.7	10.5	(29)
(30)		MDCB	8.9	11.0	18.6	20.9	26.1	28.7	29.7	28.0	26.9	22.1	16.3	12.0	(30)	
(31)		5%	WB	4.7	5.4	12.1	14.8	19.7	22.4	23.9	23.1	21.5	17.4	11.8	7.3	(31)
(32)		MDCB	6.1	7.2	15.0	19.5	24.6	27.5	28.6	26.8	25.1	20.1	14.3	8.6	(32)	
(33)		10%	WB	2.4	2.6	8.5	12.9	18.4	21.4	22.9	22.1	20.3	15.4	9.7	4.5	(33)
(34)		MDCB	3.7	4.4	11.6	17.6	22.7	26.0	27.2	25.8	23.8	18.6	12.6	6.2	(34)	
(35)	Mean Daily Temperature Range	MDBR	7.0	7.8	9.3	11.0	10.8	10.5	10.7	10.2	10.8	9.6	7.8	6.4	(35)	
(36)		5% DB	MCDBR	8.9	10.4	13.4	13.9	12.8	12.0	12.1	11.5	12.7	12.6	10.6	9.2	(36)
(37)		MCWBR	7.6	8.5	8.9	8.1	6.8	5.4	5.4	5.3	5.9	7.3	7.9	7.9	(37)	
(38)		5% WB	MCDBR	8.3	9.9	11.9	12.2	11.1	10.3	10.2	9.3	10.4	10.7	9.8	8.5	(38)
(39)		MCWBR	7.6	8.8	8.6	8.5	6.6	5.4	5.3	5.1	5.8	7.5	8.1	8.0	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.295	0.358	0.348	0.390	0.424	0.443	0.430	0.438	0.389	0.350	0.317	0.303	(40)	
(41)		taud	2.369	2.146	2.340	2.250	2.197	2.187	2.231	2.203	2.319	2.427	2.467	2.397	(41)	
(42)		Ebn at Noon	846	824	895	878	853	835	841	822	839	835	817	802	(42)	
(43)		Edh at Noon	81	119	111	131	142	144	137	136	113	90	74	73	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.39	2.24	3.52	4.62	5.36	6.06	6.16	5.30	4.28	2.72	1.68	1.14	(44)	
(45)		RadStd	0.13	0.31	0.36	0.43	0.52	0.40	0.28	0.28	0.39	0.28	0.24	0.13	(45)	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (61 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page